Sequences

9 March 2024

Problem 1. Prove that if an arithmetic sequence of positive integers contains a square, then it contains infinitely many squares.

Problem 2. Prove that one can eliminate some terms of an arithmetic sequence of positive integers in such a way that the remaining terms form a geometric sequence.

Problem 3. Does there exist an infinite non-constant arithmetic sequence whose terms are all prime numbers?

Problem 4. If the set of positive integers is partitioned into finitely many arithmetic sequences, prove that at least one of these arithmetic sequences has the property that its first term is divisible by its common difference.

Problem 5. If the set of positive integers is partitioned into n arithmetic sequences with common differences $d_1, d_2, \ldots, d_n$ prove that

$$\frac{1}{d_1} + \frac{1}{d_2} + \ldots + \frac{1}{d_n} = 1.$$

Problem 6 (APMO 2013 Problem 3)**.** For $2k$ real numbers $a_1, a_2, \ldots, a_k, b_1, b_2, \ldots, b_k$, define the sequence of numbers $\{X_n\}$ by

$$X_n = \sum_{i=1}^{k} \lfloor a_i n + b_i \rfloor \quad \text{for } n = 1, 2, \ldots.$$

If the sequence $\{X_n\}$ forms an arithmetic sequence, show that $\sum\limits_{i=1}^{k} a_i$ must be an integer

Problem 7. Prove that, for every positive integer $n \geqslant 3$, one can find an arithmetic sequence of n integers such that all of them are nontrivial powers of some integers, but one cannot find an infinite arithmetic sequence with this property.

Problem 8. Let $a_1, a_2, a_3, \ldots$ be an arithmetic sequence such that a_1^2, a_2^2 and a_3^2 are also terms of the sequence. Prove that the terms of this sequence are all integers.

Problem 9 (China MO 2023 Problem 1)**.** Let $\{a_n\}$ and $\{b_n\}$ be sequences of positive numbers satisfying the following

$$a_{n+1} = a_n - \frac{1}{1 + \sum\limits_{i=1}^{n} \frac{1}{a_i}}, \quad b_{n+1} = b_n + \frac{1}{1 + \sum\limits_{i=1}^{n} \frac{1}{b_i}},$$

for $n = 1, 2, \ldots$.

(1) If $a_{100} b_{100} = a_{101} b_{101}$, find the value of $a_1 - b_1$.

(2) If $a_{100} = b_{99}$, determine which one is larger, $a_{100} + b_{100}$ or $a_{101} + b_{101}$?

Problem 10 (IMO 2009 Problem 3)**.** Suppose that $s_1, s_2, s_3, \ldots$ is a strictly increasing sequence of positive integers such that the subsequences

$$s_{s_1},\ s_{s_2},\ s_{s_3}, \ldots \quad \text{and} \quad s_{s_1+1},\ s_{s_2+1},\ s_{s_3+1}, \ldots$$

are both arithmetic sequences. Prove that $s_1, s_2, s_3, \ldots$ is itself an arithmetic sequence.

Problem 11 (IMO 2003 Problem 5)**.** Let $x_1, x_2, \ldots, x_n$ be real numbers such that $x_1 \leqslant x_2 \leqslant \cdots \leqslant x_n$. Prove that

$$\left(\sum_{i=1}^{n}\sum_{j=1}^{n}|x_i - x_j|\right)^2 \leqslant \frac{2(n^2-1)}{3}\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - x_j)^2,$$

where equality holds if and only if $x_1, x_2, \ldots, x_n$ form an arithmetic sequence.

Problem 12 (China TST 2019 Test 1 Problem 2)**.** Fix a positive integer $n \geqslant 3$. Does there exist infinitely many sets S of $2n$ positive integers $\{a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n\}$, such that $\gcd(a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n) = 1$, and $\{a_1, a_2, \ldots, a_n\}$, $\{b_1, b_2, \ldots, b_n\}$ are arithmetic sequences with $\prod_{i=1}^{n} a_i = \prod_{i=1}^{n} b_i$?

Problem 13 (USAMO 2023 Problem 5)**.** Let $n \geqslant 3$ be an integer. We say that an arrangement of the numbers $1, 2, \ldots, n^2$ in a $n \times n$ table is *row-valid* if the numbers in each row can be permuted to form an arithmetic sequence, and *column-valid* if the numbers in each column can be permuted to form an arithmetic sequence.

For what values of n is it possible to transform any row-valid arrangement into a column-valid arrangement by permuting the numbers in each row?